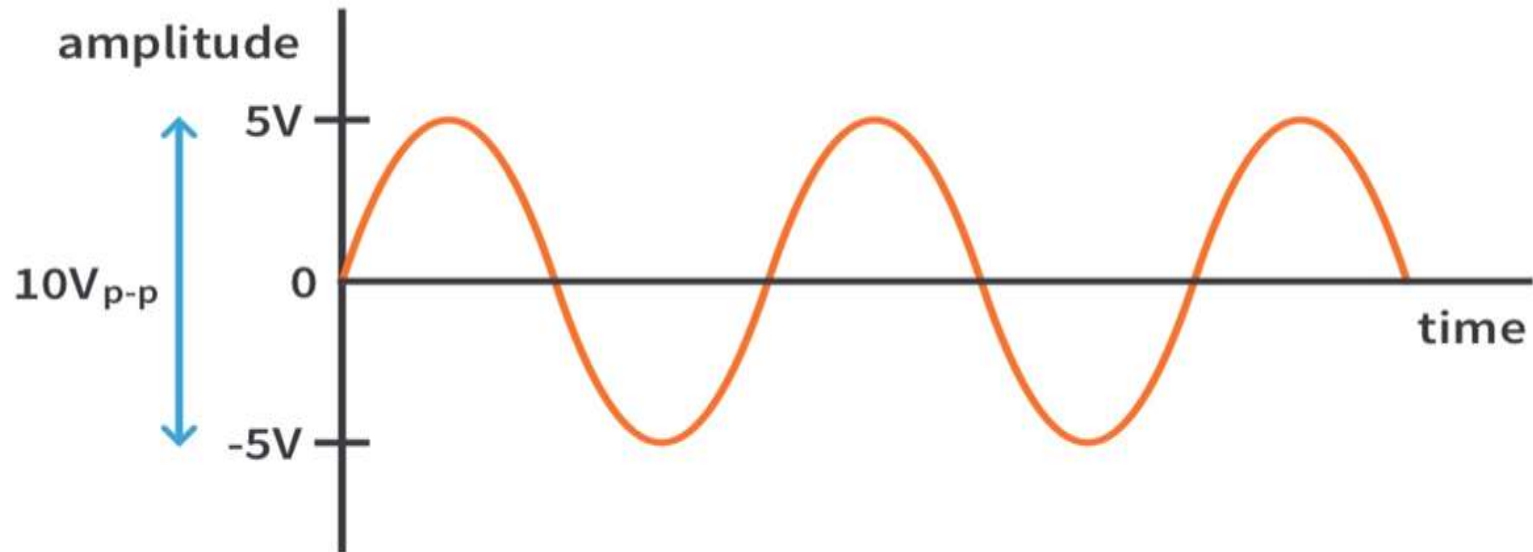


Unit-1

Introduction

Analog System

- The system Which can Process analog quantities (continuous data) is called an Analog System.
- Analog system is operated by measuring rather than counting.
- Eg. Speedometer, Thermometer, Voltmeter etc.



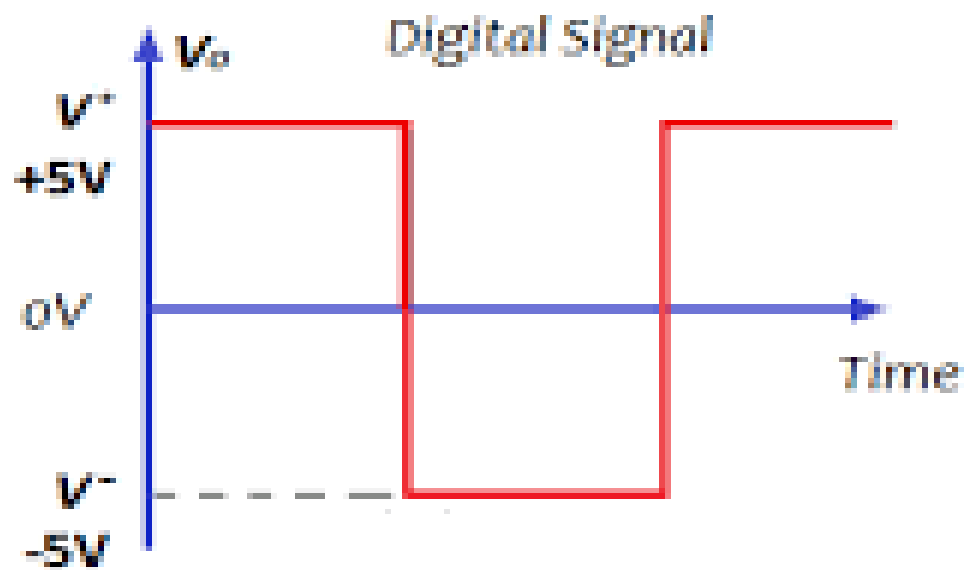
Characteristics of analog signal

- Based on continuous varying data
- Measure only natural or physical values
- Used for special purpose
- Generally, no storage facility because they work on real time basis.
- Accuracy of these types of computer is very less because of noise and filtering facility.
- Output of these signal is not known by general public because they are in form of wave lines, curved lines or graph.

Digital System

- The system which works on discrete data (discontinuous data, binary system or 0 and 1) is called digital system.
- Binary system is such system of numbering in which only 2 digits are used 0 and 1. where 0 represents either OFF, False, No and 1 represents ON, True, Yes etc. So, the basic principle of this system is either present or absent of electrical pulses in the signal.

Digital signal



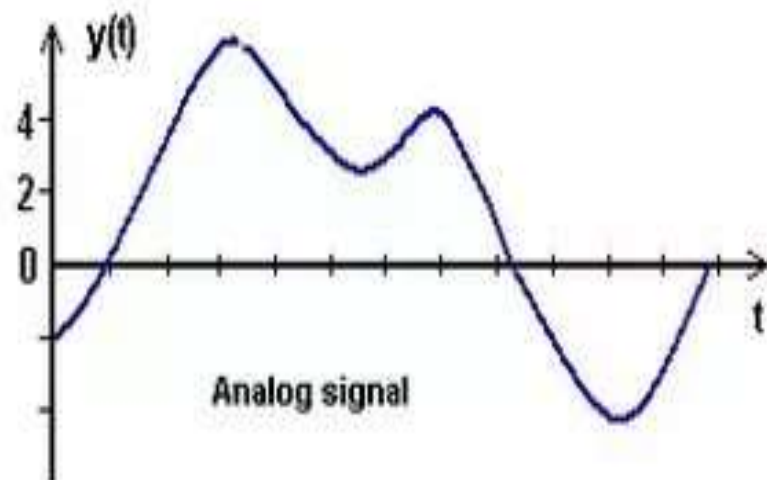
Characteristics of Digital Signal

- Based on discrete data which are not continuous with time
- Based on principle of logic 0 and 1 (high and low voltage)
- Used for general purpose.
- They are more reliable because of less noise and filtering facility.
- It has large memory capacity because the calculations are to be stored internally for future use and re-programming.
- It is multipurpose and programmable. So, it is of high cost and faster processing

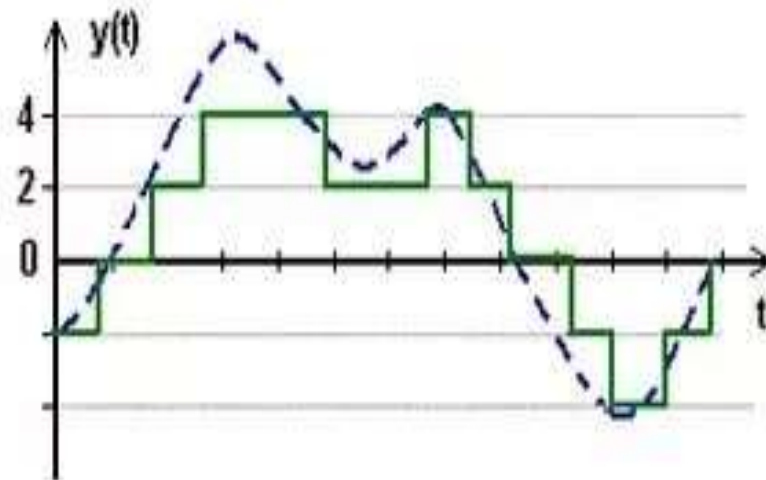
Advantages of digital signal

- Digital data can be transmitted more efficiently and is more reliable than analog data
- Digital data can be stored in memory.
- Information storage is easy
- It is less affected from the noise
- The transmission is more secure with encryption
- Easier to design. Exact value of voltage or current are not important, only the range (HIGH or LOW) in which they fall

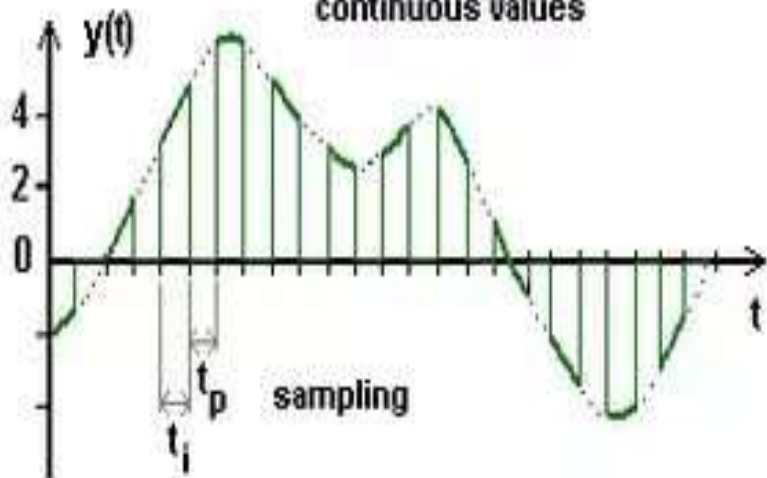
continuous-time
continuous values



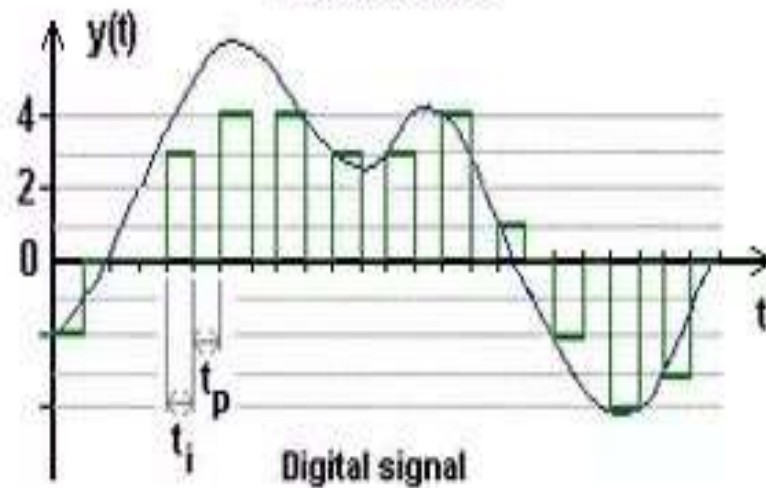
continuous-time
discrete-value



discrete-time
continuous values



discrete-time
discrete-value



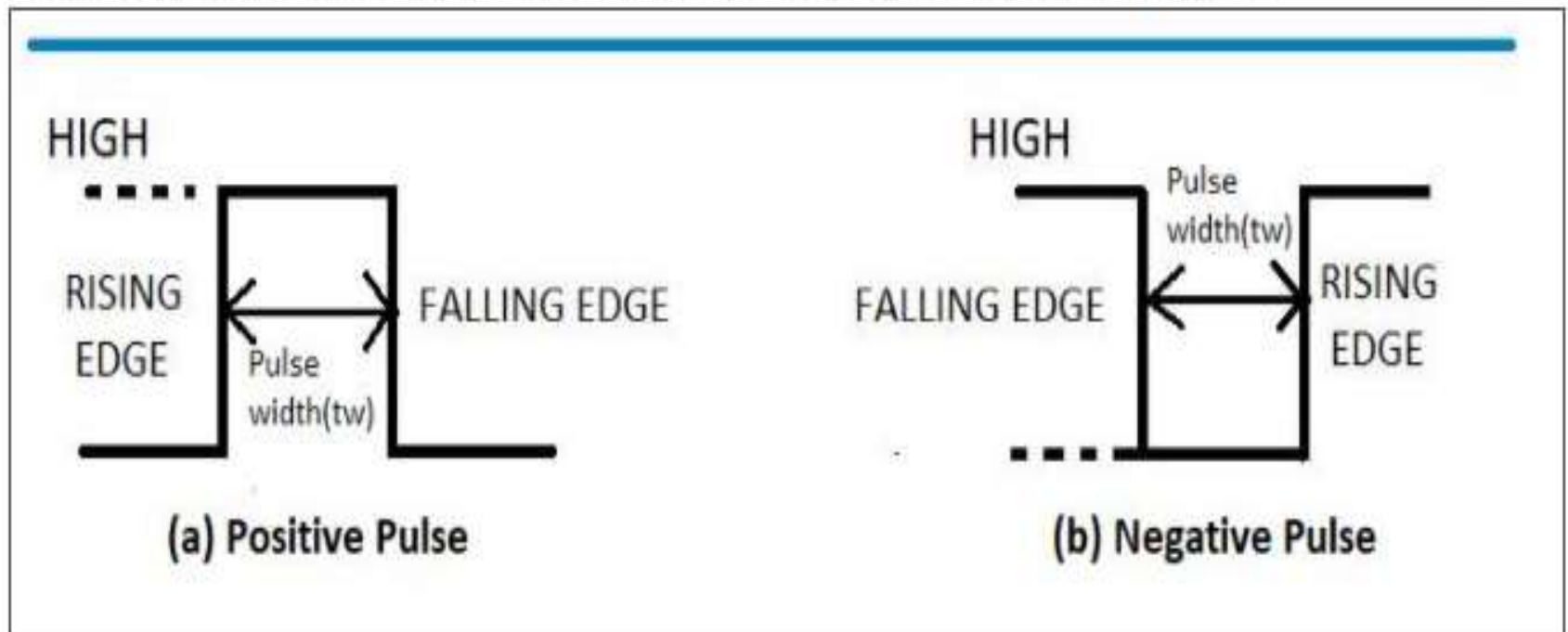
Difference between Analog and Digital Signal

S.No.	Analog Signal	Digital signal
1.	These signal works with natural or physical values.	These signal works with digits.
2.	It works upon continuous data.	It works upon discrete data
3.	It usually in the form of sine wave.	It is usually in the form of square wave.
4.	Its accuracy is low	Its accuracy is high
5.	Output is continuous	Result are obtained after complete computation
6.	No any or smaller storage capacity	Larger storage capacity (memory)
7.	Lower cost compared to digital systems	Higher cost compared to analog system
8.	It consume less bandwidth	It consume more bandwidth
9.	Example: Speedometer, Thermometer, Voltmeter etc.	Example: Smart Watch, Desktop computer etc.

Digital Waveforms

Digital waveforms are made of voltage levels that are HIGH AND LOW and also they contain a series of pulses.

Look at the diagrams below to know more about the specific terms in a pulse:



Characteristics of Digital waveform

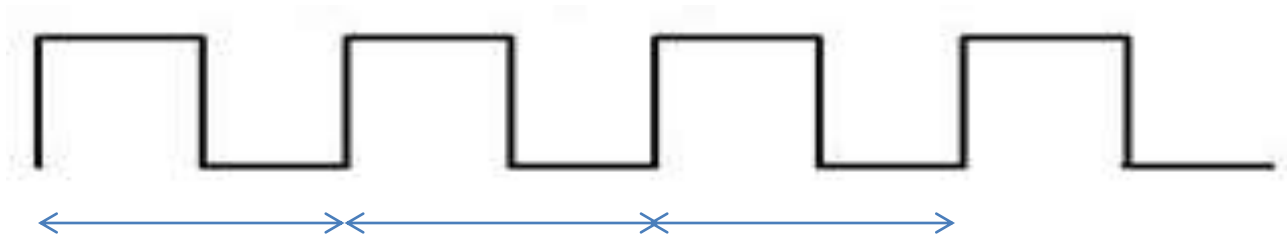
- Frequency:

Number of pulses in 1 second.

It is the rate at which signal repeats itself.

Its unit is Hz

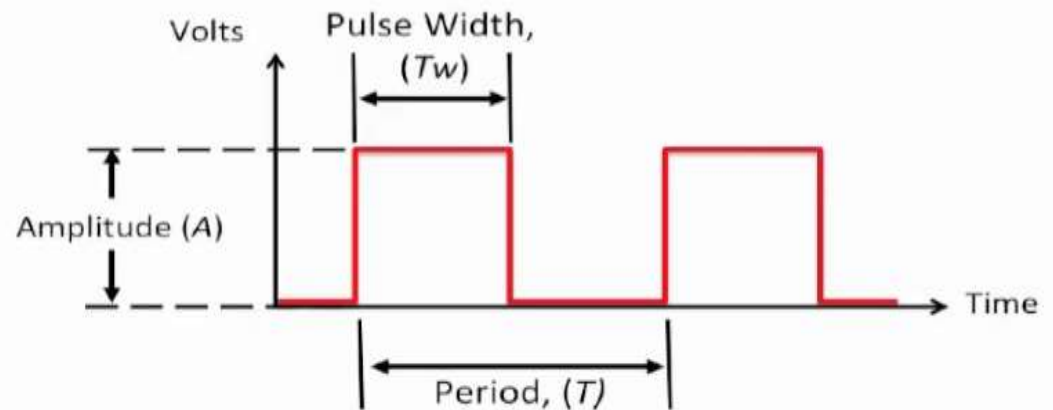
$$f=1/T$$



- Duty cycle:

It is the ratio of the pulse width to the period and is expressed in percentage.

Duty Cycle = $\frac{tw}{T} * 100\%$ where, tw is the pulse width and T is the period.



- Amplitude:

The maximum displacement of a pulse.

- Time period :

It is the time over which a signal repeats itself.

- Rise Time:

The time required for the pulse to go from its LOW level to HIGH level is called rise time (t_r). In practice, it is common to measure rise time from 10% of the pulse amplitude to 90% of the pulse amplitude.

- Fall time(t_f):

It is the time required for the transition from high level to low level. It is the time from 90% to 10% of the pulse amplitude.

Digital logic

- Digital or Boolean, logic is the fundamental concept underpinning all modern computer systems. Put simply, it's the system of rules that allow us to make extremely complicated decisions based on relatively simple "yes/no" questions.
- It is the representation of signals and sequences of a digital circuit through numbers.

Digital circuitry

- Digital logic circuits can be broken down into two subcategories- **combinational** and **sequential**.
- Combinational logic changes "instantly"- the output of the circuit responds as soon as the input changes (with some delay, of course, since the propagation of the signal through the circuit elements takes a little time).
- Sequential circuits have a clock signal, and changes propagate through stages of the circuit on edges of the clock.

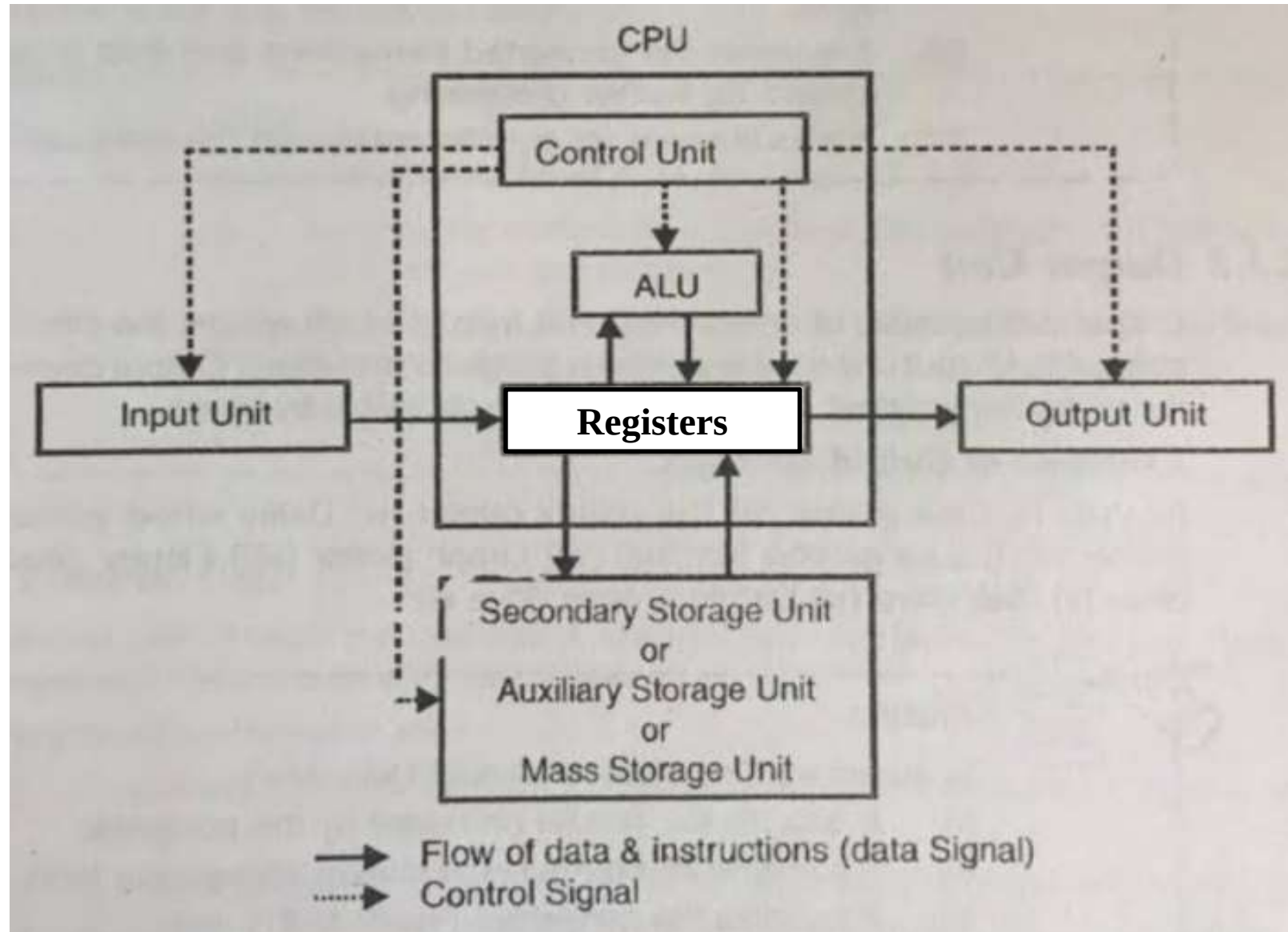
Moving and storing digital information

- Flip-flops
- Registers
- Memory Devices

Digital Operations

- Counters
- ALU
 - Arithmetic Operations
 - Logic Operations
- Adders and Subtractors
- Comparator

Digital Computer



Input Unit

- Input unit connects the external environment with internal computer system. It provide data and instructions to the computer system.
- Commonly used input devices are keyboard, mouse, joy stick, camera, scanner etc.
- Input unit accept data and instructions from the outside environment and convert it into machine language.

Output unit

- Output unit are the peripheral devices that provide the output of the calculations made by the computer system to the user. Some output devices are printers, monitors, speaker etc.
- Types:

Hardcopy output devices

-printer

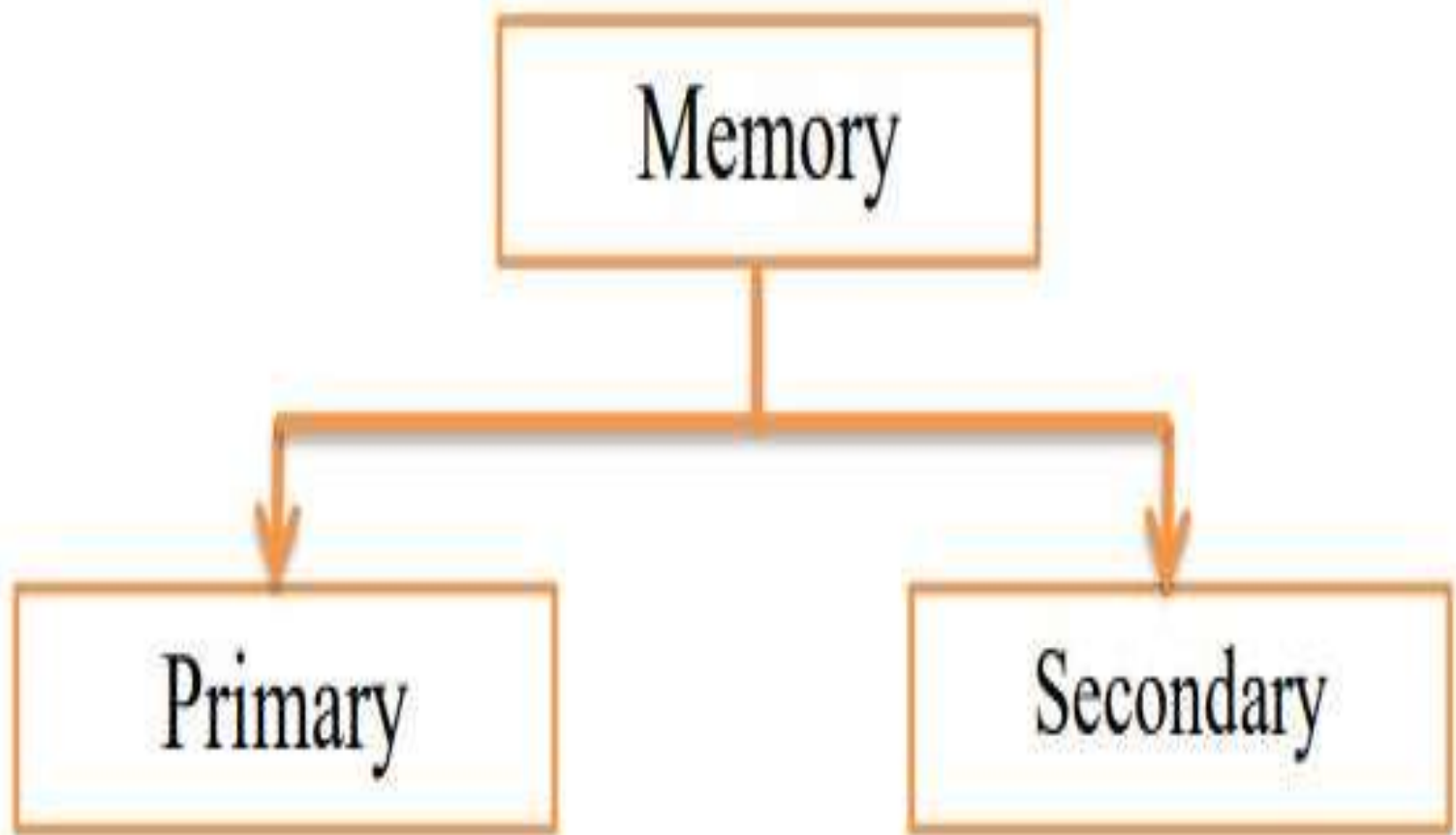
Softcopy output devices

-monitor, speaker computers, laptops, smartphones etc.

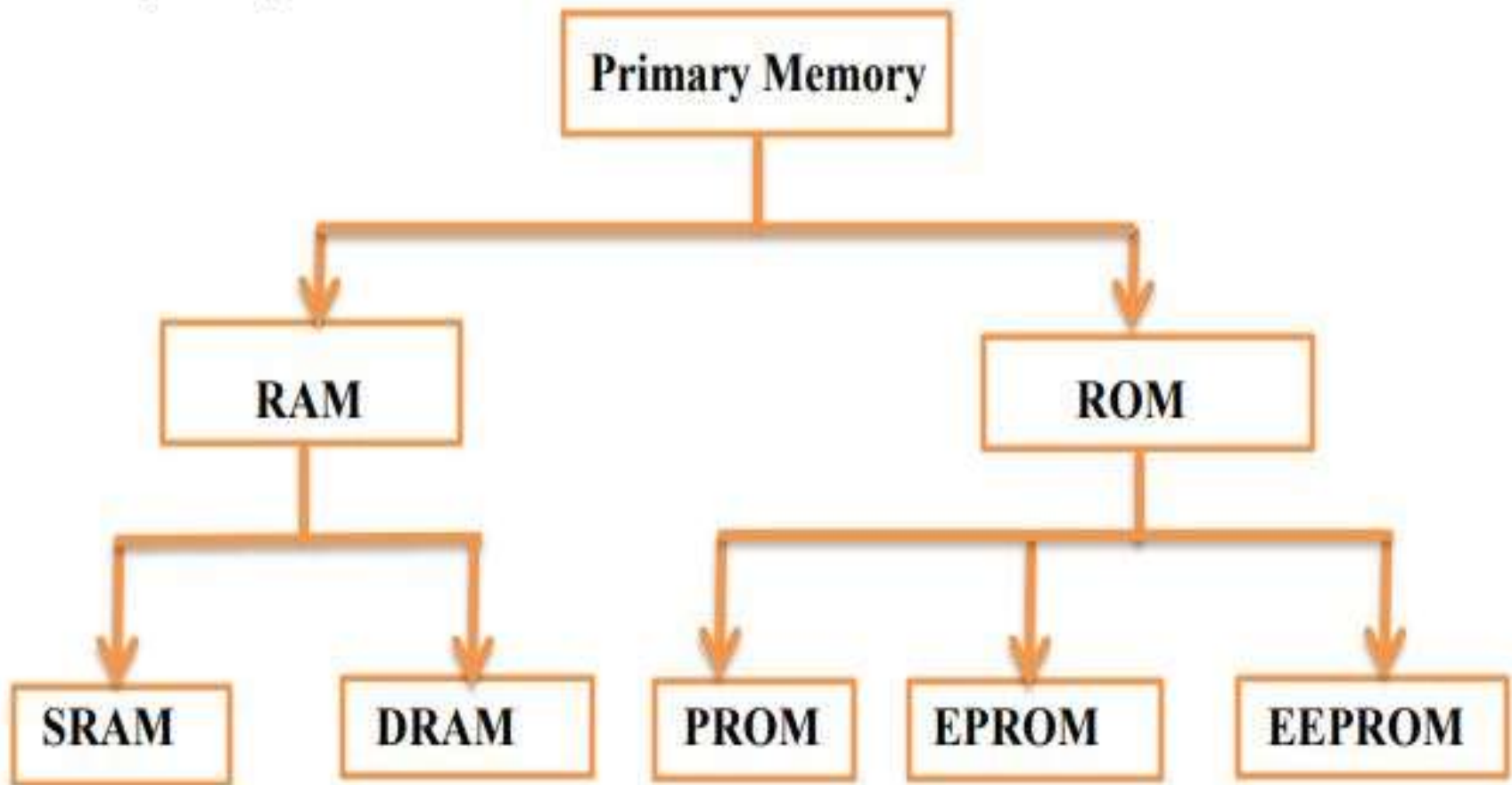
Central Processing Unit

- CPU is called brain of a computer
- It is responsible for all function and processes
- It is also responsible for controlling the operations of other units of computer system.
- The CPU consists
 - Arithmetic and logical unit
 - Control unit
 - Register

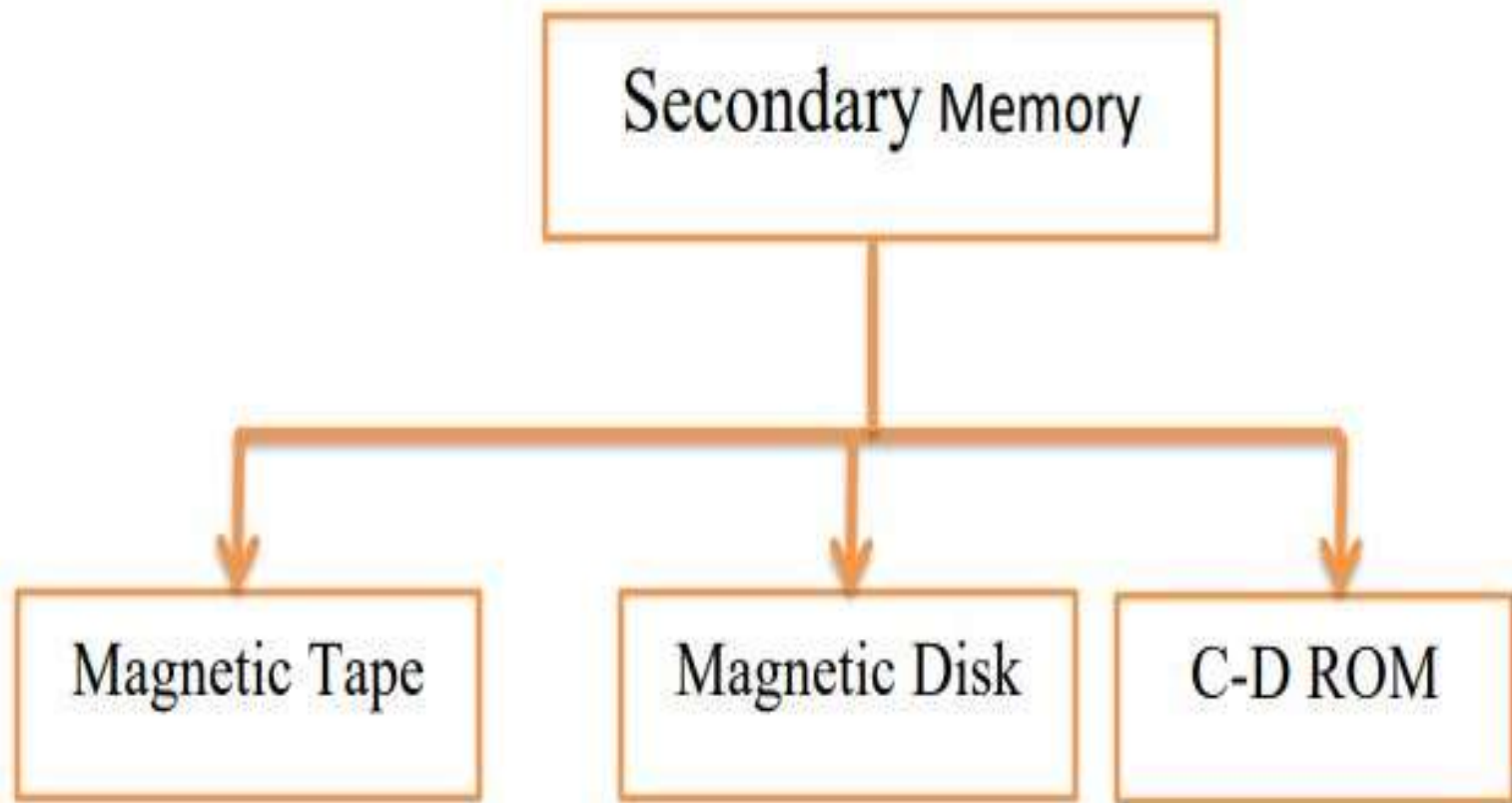
Memory Unit



Primary memory



Secondary memory



Digital Integrated Circuit

- An integrated circuit (also referred to as an IC, a chip, or a microchip) is a set of electronic circuits on one small flat piece (or "chip") of semiconductor material, usually silicon.
- A digital IC is constructed by an interconnection of resister, transistors and perhaps small capacitor, all of which have been formed on the surface of silicon wafer.
- The entire circuit resides on a tiny piece of semiconductor material called a chip.
- This chip comprises of resisters, transistors, logic gates, capacitors.

Levels of Integration

Small-scale integration (SSI):

no of gates less than 10

- Simple logic functions, basic storage elements etc.

Medium-scale integration (MSI):

10–100 gates

- 4-bit adders, 8-to-1 multiplexers, counters, etc.

Large-scale integration (LSI):

100 to a few 1000 gates

- Programmable logic array (PLA), PROM, small processors, etc.

Very large-scale integration (VLSI):

several thousand to millions of gates

- Processors, large memory array, etc.

Ultra large-scale integration (ULSI):

no of gates > 10 million

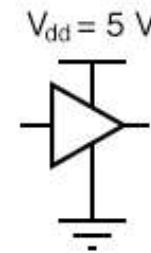
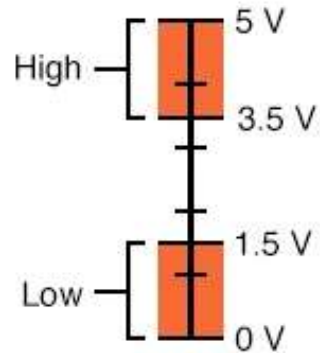
- Advanced processors, system chips, etc.

Digital Logic Families

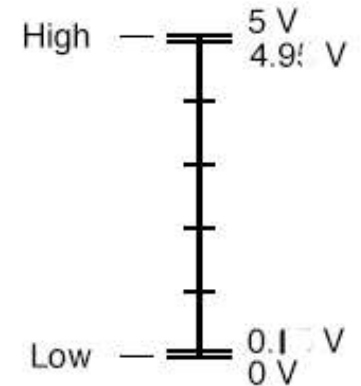
- RTL (Resistor-Transistor Logic):
- DTL(Diode-Transistor Logic):
- TTL(Transistor-Transistor Logic):
- ECL (Emitter-Coupled Logic):
- MOS (Metal-Oxide Semiconductor):
- CMOS (Complementary Metal-Oxide Semiconductor):

Digital IC Signal Levels

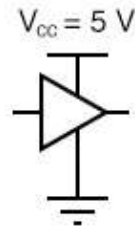
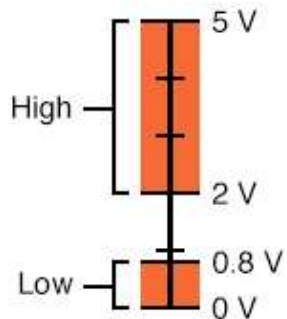
Acceptable CMOS Gate
Input Signal Levels



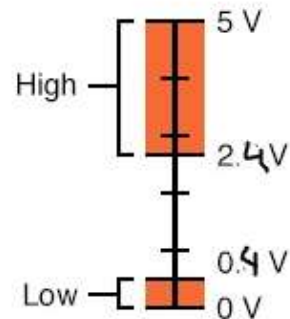
Acceptable CMOS Gate
Output Signal Levels



Acceptable TTL Gate
Input Signal Levels



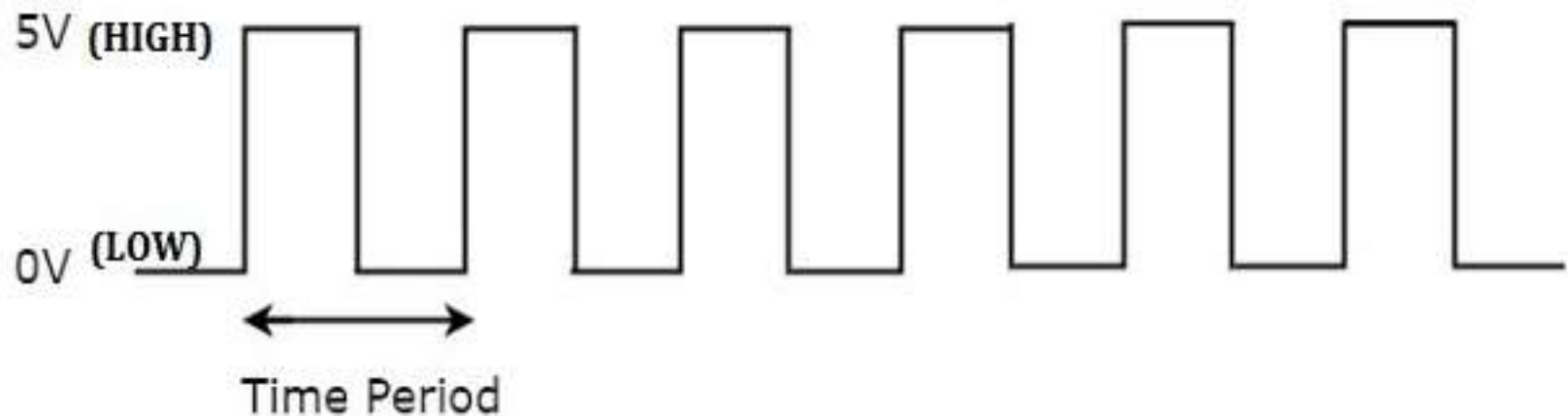
Acceptable TTL Gate
Output Signal Levels



Clock Waveform

Clock is a square waveform switching from high to low indefinitely at fixed intervals often employed in digital electronic circuits to synchronize the function of each circuit element.

Here is how a clock looks



Clock has fixed time period and generally in most applications duty cycle is 50% i.e. for half of the time period its value is high and for other half it is low.

End of Unit 1

THANK YOU